

**CONCISE DESCRIPTION OF RELEVANCE: THIRD-PARTY SUBMISSION UNDER 37 CFR § 1.290**

*U.S. Application No. 19/318,450 (Publication No. US 2026/0070232 A1)*

**Document 9: U.S. Patent Application Publication No. US 2025/0319560 A1 (“Document 9”)**

*First Named Inventor:* David H. Morton

*Publication Date:* October 16, 2025

*Title:* Magnetic coupling device with at least one of a sensor arrangement and a degauss capability

This document discloses a magnetic system for moving a ferromagnetic workpiece, which includes magnetic coupling devices, magnetic field sensors, and a calibration module. Its potential relevance to the claims of the above-identified application is set forth below.

| Claim language (US 2026/0070232 A1)                                                                                                                                                                                                                 | Relevant disclosure of this document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Claim 1: “A magnetic gripper for gripping a ferromagnetic workpiece, said magnetic gripper comprising”                                                                                                                                              | The reference discloses a "magnetic system for moving a ferromagnetic workpiece" which includes "magnetic coupling devices" configured to "engage the ferromagnetic workpiece".<br>“A magnetic system for moving a ferromagnetic workpiece, the magnetic system comprising: a machine including a base and an arm having a plurality of arm segments, a first end of the plurality of arm segments being opposite the base; a ...” <ul style="list-style-type: none"><li>• Claim 1</li></ul>                                                                                                                          |
| Claim 1: “a magnet which is arranged in the housing and which is displaceable along a displacement direction between a gripping position for gripping the ferromagnetic workpiece and a release position for releasing the ferromagnetic workpiece” | The reference discloses that the "at least first permanent magnet is moveable relative to the housing".<br>“The magnetic system of claim 1, wherein the at least first permanent magnet is moveable relative to the housing.” <ul style="list-style-type: none"><li>• Claim 1</li></ul>                                                                                                                                                                                                                                                                                                                               |
| Claim 1: “a number of pole shoes attached to the housing, wherein each pole shoe has a workpiece contact surface and is designed to guide a magnetic field portion of the magnet to the workpiece contact surface”                                  | The reference discloses "a plurality of workpiece engagement surfaces supported by the housing and magnetically coupled to the magnetic flux source, the plurality of workpiece engagement surfaces adapted to contact the ferromagnetic workpiece".<br>“a plurality of workpiece engagement surfaces supported by the housing and magnetically coupled to the magnetic flux source, the plurality of workpiece engagement surfaces adapted to contact the ferromagnetic workpiece, a first workpiece engagement surface of the plurality of workpiece ...” <ul style="list-style-type: none"><li>• Claim 1</li></ul> |

| Claim language (US 2026/0070232 A1)                                                                                                                                                                | Relevant disclosure of this document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Claim 8 (The magnetic gripper according to claim 1, wherein the number of pole shoes includes a first pole shoe and a second pole shoe)</p>                                                     | <p>The reference discloses "a first workpiece engagement surface of the plurality of workpiece engagement surfaces corresponding to a north pole of the magnetic coupling device and a second workpiece engagement surface of the plurality of workpiece engagement surfaces corresponding to a south pole".</p> <p>“a first workpiece engagement surface of the plurality of workpiece engagement surfaces corresponding to a north pole of the magnetic coupling device and a second workpiece engagement surface of the plurality of workpiece engagement surfaces corresponding to a south pole ...”</p> <ul style="list-style-type: none"> <li>• Claim 1</li> </ul> |
| <p>Claim 9 (The magnetic gripper according to claim 1, wherein the magnet is displaceable electrically, pneumatically, or mechanically between the gripping position and the release position)</p> | <p>The reference discloses "an actuator operably coupled to the magnetic platter, the actuator operable to translate the magnetic platter relative to the housing".</p> <p>“further comprising an actuator operably coupled to the magnetic platter, the actuator operable to translate the magnetic platter relative to the housing.”</p> <ul style="list-style-type: none"> <li>• Claim 1</li> </ul>                                                                                                                                                                                                                                                                   |
| <p>Claim 10 (the magnetic gripper has a workpiece sensor for detecting the ferromagnetic workpiece)</p>                                                                                            | <p>The reference discloses "the controller configured to determine a status of each of the first magnetic coupling device and the second magnetic coupling device relative to the ferromagnetic workpiece".</p> <p>“the controller configured to determine a status of each of the first magnetic coupling device and the second magnetic coupling device relative to the ferromagnetic workpiece.”</p> <ul style="list-style-type: none"> <li>• Claim 1</li> </ul>                                                                                                                                                                                                      |
| <p>Claim 14 (A magnetic gripper system, the magnetic gripper system comprising)</p>                                                                                                                | <p>The reference discloses "A magnetic system for moving a ferromagnetic workpiece, the magnetic system comprising: a machine including a base and an arm having a plurality of arm segments".</p> <p>“A magnetic system for moving a ferromagnetic workpiece, the magnetic system comprising: a machine including a base and an arm having a plurality of arm segments ...”</p> <ul style="list-style-type: none"> <li>• Claim 1</li> </ul>                                                                                                                                                                                                                             |